

Shengfan Cao

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Research Interests

My research focuses on integrating safety constraints, optimization, and learning to enable reliable long-horizon decision-making for embodied AI systems under uncertainty.

Education

University of California, Berkeley

Aug 2022 — May 2027

Ph.D. Candidate in Mechanical Engineering

- **PI:** Francesco Borrelli
- **Coursework:** Advanced Control Theory and Systems, Model Predictive Control, Nonlinear Systems, Convex Optimization, Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing.

Tsinghua University

Aug 2017 — Jun 2022

B.S. in Mechanical Engineering; B.A. in Japanese Language and Literature

- **PI:** Chuxiong Hu (Mechanical Engineering); Rong Zhao (Japanese Language and Literature)
- **Thesis:** Path Generation for Rapid Filling of Planar Areas with Optimized Parallel Curves

Industry Experience

DiDi Research America

May 2026 — Aug 2026

Motion Planning Engineer, Intern

- Integrated a DDP smoother into the training and preprocessing pipeline of an end-to-end planning model trained on **1.3B human-driving frames**, improving comfort and efficiency in data-based simulation.
- Reformulated trajectory cleanup as Bayesian MAP estimation and shipped an augmented-Lagrangian DDP denoiser, reducing jerk and lateral-jerk discomfort rates to **0–4% from 8–39%** while keeping ADE/FDE competitive.
- Re-architected the trajectory-optimization core of a large legacy C++ production planner as an efficient augmented-Lagrangian DDP solver, improving solve speed by $\sim 6\times$ over an IPOPT baseline.
- Designed an elastic constraint-relaxation scheme with L_1 -penalized slacks and shipped it onboard as a no-op-safe relaxable augmented-Lagrangian constraint type.
- Coordinated design and code reviews across ML-planner and path-planner teams; evaluated changes on real driving scenarios across safety, compliance, comfort, and efficiency metrics; and participated in weekly road-test triage.
- **Tech:** C++17, Python, CasADi, Acados/HPIPM/BLASFEO, IPOPT, Eigen, pybind11, ROS 1, Bazel.

Major Research Experience

Trusted Autonomy for Rapidly Prototyped Uncrewed Ground Vehicles, Project Lead

Sep 2024 — Present

- Led a rapidly deployable, real-time autonomy stack for unseen terrain using stereo infrared cameras, GPS, IMU, and motor telemetry; fused V-SLAM/GPS/IMU with an EKF to produce live terrain, obstacle, and uncertainty maps.
- Built an obstacle-understanding pipeline aligning 3D point clouds with semantically segmented RGB images, enabling online semantic mapping and real-time replanning as terrain and obstacles were discovered.
- Implemented RRT*, A*, and D* Lite over a continuously updated semantic world representation, and deployed the PyTorch/ROS 2 stack on Jetson Orin under onboard compute constraints.
- Built a Python/SQLite layer for querying logged ROS 2 autonomy data and collaborated with NIWC Pacific human-factors researchers through regular technical reviews.
- **Tech:** PyTorch, ROS 2, SQL, V-SLAM, EKF, YOLO, semantic mapping, OpenCV, CasADi, Jetson Orin.

Safe Diffusion Policies for Robot Learning, Principal Researcher*Mar 2026 — Present*

- Investigating inference-time safety mechanisms for diffusion policies in long-horizon robot decision-making.
- Built a PyTorch training and evaluation framework for Diffusion Transformer policies across navigation and manipulation domains.
- Designed experiments on safety-performance tradeoffs and closed-loop consistency, demonstrating promising path-planning results and identifying challenges in receding-horizon control and manipulation.
- **Tech:** PyTorch, Diffusion Transformers, robot learning, path planning, Gymnasium.

State-Conditional Adversarial Learning, Co-author*2025 — 2026*

- Co-authored an off-policy visual-domain-transfer method for end-to-end imitation learning that aligns source and target latent distributions conditional on vehicle state using a discriminator-based conditional-KL estimator.
- Evaluated the method in visually diverse BARC-CARLA driving environments, demonstrating robust transfer and strong sample efficiency with scarce, expert-free target-domain data.
- **Tech:** PyTorch, CARLA, domain adaptation, imitation learning, adversarial learning.

Sampling-Based Constrained Policy Optimization, Principal Researcher*Sep 2025 — Dec 2025*

- Introduced a scalable, training-time constrained optimization framework for large neural policies, enabling safety guarantees without test-time shielding or policy rollback.
- Formulated policy optimization as a convex projection in parameter space, with theoretical guarantees that projected updates preserve objective improvement.
- Derived a sufficient stability condition linking weight updates to closed-loop constraint satisfaction, and embedded it directly into the optimization pipeline.
- Empirically demonstrated complete rejection of harmful supervision and safe performance gains in regression and imitation learning under adversarial experts.
- Oral presentation at IFAC World Congress 2026.
- **Tech:** PyTorch, Convex Optimization, Reinforcement Learning, Safe Learning.

Safe Imitation Learning at Handling Limits, Principal Researcher*Nov 2024 — Feb 2025*

- Designed a constraint-aware imitation learning framework incorporating an actor-critic structure for explicit safety modeling.
- Improved training efficiency by 150% and reduced collision rates by 60% at the vehicle's dynamic handling limits in CARLA simulation.
- Integrated reachability-based safety filters into the training pipeline, enabling safe policy generalization.
- Oral presentation at IEEE IROS 2025 for this work: "*A Simple Approach to Constraint-Aware Imitation Learning with Application to Autonomous Racing*".
- **Tech:** PyTorch, CasADi, CARLA, RL, MPC, optimization, safety-aware learning.

Vision-based End-to-end Control for Racing, Principal Researcher*Nov 2023 — May 2024*

- Developed a CNN-based end-to-end controller for high-speed autonomous racing using RGB camera input and velocity feedback.
- Trained using imitation learning from MPCC expert trajectories and deployed on a 1:10 Jetson-powered vehicle with onboard ROS stack.
- Designed and executed systematic policy evaluation in CARLA under various conditions (e.g., weather, lighting), measuring success rates and failure modes across diverse initial conditions to assess robustness near dynamic handling limits.
- Achieved long rollouts (80 laps at high speed) without constraint violation and improved consistency in performance compared to traditional SLAM-based pipelines across 10+ field tests.
- **Tech:** PyTorch, CasADi, ROS, OpenCV, SLAM, RL, NVIDIA Jetson, real-time control.

Additional Project & Research Experience

Goal-Conditioned Learning MPC, Independent Research

May 2023 — Sep 2023

- Explored a goal-conditioned, data-driven MPC pipeline using feasible closed-loop trajectories to learn transition costs between feasible states via imitation learning.
- Investigated a two-layer hierarchical architecture: short-horizon reference generation in a planning layer, tracked by MPC in an execution layer.
- Demonstrated improved long-horizon efficiency in simulation compared to a single-layer MPC baseline.
- **Tech:** Gurobi, CasADi, MPC, Python.

Autonomous Rover for Solar Panel Fields (Terabase), Team Member

May 2023 — Aug 2023

- Co-implemented a perception stack on ROS 1 to identify object locations and types around the rover.
- Assisted in connecting perception outputs with an MPC planner and low-level controller on ROS 2 to enable evasive maneuvers.
- Debugged on the physical rover and delivered a field demo in Davis.
- **Tech:** ROS 1/2, MPC, YOLO, Python.

Corner Smoothing for Additive Manufacturing Toolpaths, Research Assistant

May 2021 — Aug 2021

- Proposed a corner smoothing method for additive manufacturing toolpaths to achieve uniform coating while constraining maximum curvature.
- Refined the original formulation into a convex optimization problem to enforce continuous curvature and even area coverage.
- Implemented an iterative solver (gradient-based) in Python and integrated it into a parallel-curve toolpath planner.
- **Tech:** Convex optimization, Python.

Path Planning for Rapid Filling of Planar Areas with Optimized Parallel Curves, Research Assistant

Mar 2021 — May 2021

- Proposed and implemented an algorithm to generate non-self-intersecting parallel curves that cover the area enclosed by an arbitrary boundary curve.
- Built a net-like linked list representation to compute normal vectors and curvature pointwise using central differences.
- Improved motion stability by converting contour-like paths into spiral-like paths using offsetting methods.
- **Tech:** Computational geometry, Python/C++.

Publications & Preprints

- Cao, S., Joa, E., & Borrelli, F. (2025). *A Simple Approach to Constraint-Aware Imitation Learning with Application to Autonomous Racing* [🔗](#). Oral presentation at IEEE IROS 2025.
- Prignoli, F., Cao, S., Falcone, P., Borrelli, F. (2025). *Real-Time Regulation-Aware Game-Theoretic Motion Planning for Head-to-Head Autonomous Racing*. Under review at IEEE TCST.
- Cao, S., Borrelli, F., & Joa, E. (2025). *Constrained Policy Optimization via Sampling-Based Weight-Space Projection* [🔗](#). Oral presentation at IFAC World Congress 2026.
- Liu, Y., & Cao, S. (2025). *State-Conditional Adversarial Learning: An Off-Policy Visual Domain Transfer Method for End-to-End Imitation Learning* [🔗](#). IFAC World Congress 2026.

Teaching

Supervisor for Undergraduate Research Program, UC Berkeley

Jan 2025 — Present

- Led a team of four undergraduate students on vision-based off-road vehicle dynamics modeling and path planning, guiding research efforts, technical documentation, and successful interim presentations to faculty and industry stakeholders.

Graduate Student Instructor, UC Berkeley

Jan 2024 — May 2024

- Led discussions 10 hours per week for 40-student sessions.
- Held office hours 2 hours per week that helped more than 20 students.
- Developed labs and exams on Python, MATLAB, and numerical methods to prepare students for advanced engineering classes.

Graduate Student Instructor, UC Berkeley

Jan 2025 — May 2025

- Developed a vehicle dynamics simulator based on dynamic bicycle model, with standard gymnasium APIs.
- Implemented a CARLA connector in the gymnasium wrapper and a shell script for quick installation. [🔗](#)

Students Mentored

Yuxiang Liu, UC Berkeley	Jan 2023 — Present
Yun Chung Chang, UC Berkeley	Jan 2025 — Present
Parham Sharafolslami, UC Berkeley	Jan 2024 — Dec 2024
Ethan Kou, UC Berkeley	Jan 2025 — May 2025
JongHoon Ock, UC Berkeley	Jan 2025 — May 2025

Academic Services

Invited Reviewer, IEEE and IFAC Venues

Apr 2025 — Present

- Reviewed seven submissions across Automatica, CDC, IFAC World Congress, IEEE CASE, and IEEE Robotics and Automation Letters.

Awards & Honors

- Scholarship for Academic Excellence (Top 10%) sponsored by AEON
- Scholarship for Art Excellence
- Scholarship for Social Services
- First Honor Roll, 8th U.S. United Young Musician International Piano Competition

Skills

- **Programming:** Python, C++17, C, MATLAB, SQL, Java, HTML.
- **Software:** PyTorch, NumPy, Pandas, OpenCV, ROS 1/2, CARLA, CasADi, Acados, Eigen, pybind11, Gymnasium, Linux/bash, Bazel, Docker, Git, Pytest, NVIDIA Visual SLAM, SolidWorks, L^AT_EX.
- **Hardware:** Raspberry Pi, NVIDIA Jetson Orin Nano, Arduino.
- **Languages:** Japanese (JLPT N1); English (fluent); Chinese (native).